

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and

Agent COURSE OUTCOME COVERED

CO5: Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)

Assessment Tool Weightage: 35%

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Code of Conduct:

I K.Neeha (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

QUESTIONS: 10 Total Marks: 50

1. Simulate a Hierarchical Reinforcement Learning (HRL) Agent for Autonomous Warehouse Navigation

(5 Marks)

Problem Statement

Design and simulate a Hierarchical Reinforcement Learning (HRL) agent for an autonomous warehouse robot. The robot must navigate through different sections of a warehouse to collect and deliver packages. Instead of learning the complete navigation task as a single problem, the task should be divided into multiple levels of decision-making.

The high-level controller selects a sub-task such as move to storage area, collect package,

move to dispatch area, or deliver package. The low-level controller performs primitive actions such as moving Up, Down, Left, or Right.

Requirements

- Represent the warehouse as a grid environment.
- Define obstacles, storage locations, and delivery locations.
- Create high-level and low-level policies.
- Use rewards for successful navigation and package delivery.
- Penalize collisions, unnecessary movements, and invalid actions.
- Simulate the robot until the package is successfully delivered.

Expected Output

The simulation should demonstrate how HRL decomposes a complex warehouse navigation problem into smaller sub-tasks and enables the robot to reach the destination efficiently.

Key Concept

HRL → High-level policy → Sub-task → Low-level policy → Primitive action → Environment

2. Implement a Hierarchical Abstract Machine (HAM) to Control a Robot Performing Sequential Tasks

(5 Marks)

Problem Statement

Implement a **Hierarchical Abstract Machine (HAM)** for a service robot that performs a sequence of tasks in an indoor environment. The robot must complete tasks such as **moving to a location, picking up an object, transporting it, and placing it at the target location.**

The HAM should represent the robot's behavior using a hierarchy of machines or control states. Each machine should determine which action or sub-task is executed based on the current state of the robot.

Requirements

- Define the robot's environment and state representation.
- Create hierarchical states for different tasks.
- Define task sequences such as:
 1. Navigate to object
 2. Pick up object
 3. Navigate to destination
 4. Place object
- Define primitive robot actions.
- Use conditional transitions between states.
- Provide rewards for successful task completion.
- Handle invalid actions and obstacles.

Example Hierarchy

Main Task

↓

Navigate to Object

↓

Move → Avoid Obstacle → Reach Object

↓

Pick Object

↓

Navigate to Destination

↓

Move → Avoid Obstacle → Reach Destination

↓

Drop Object

Expected Output

The robot should execute the tasks in the correct order using the hierarchical machine and successfully complete the overall objective.

Key Concept

HAM provides structured control over an RL agent by restricting and organizing the possible actions through a hierarchy of states and transitions.

3. Develop the MAXQ Value Decomposition Algorithm for a Taxi Navigation

Environment

(5 Marks)

Problem Statement

Develop a **MAXQ Value Decomposition** algorithm for a taxi navigation environment. The taxi agent must pick up a passenger from one location and transport the passenger to the required destination.

Instead of representing the entire taxi task using a single value function, MAXQ should decompose the problem into smaller subtasks such as:

- Navigate to passenger
- Pick up passenger
- Navigate to destination
- Drop passenger

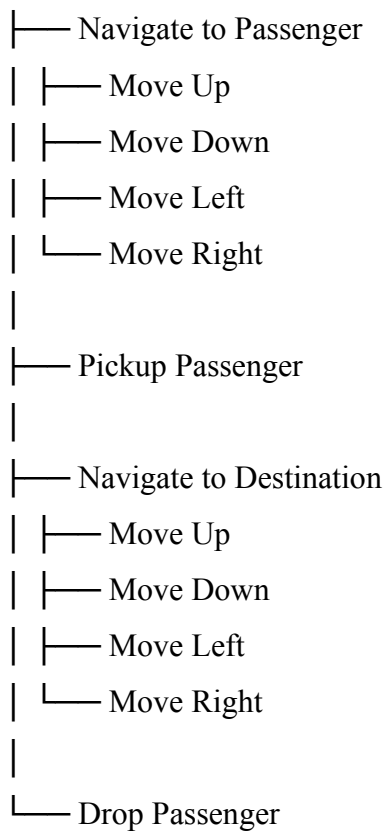
Requirements

- Create or use a grid-based taxi environment.
- Define taxi, passenger, and destination states.
- Define primitive actions such as:
 - Up
 - Down

- Left
- Right
- Pick-up
- Drop-off
- Divide the complete task into hierarchical subtasks.
- Calculate values for individual subtasks.
- Combine subtask values to estimate the overall task value.
- Provide positive reward for successful delivery.
- Penalize unnecessary movement and illegal pickup/drop-off operations.

MAXQ Structure

Taxi Delivery Task



Expected Output

The algorithm should learn appropriate values for the subtasks and gradually improve the taxi's ability to complete passenger delivery with fewer steps.

Key Concept

MAXQ decomposes the value of a complex task into values associated with smaller subtasks, making hierarchical decision-making easier to learn.

4. Simulate a Meta-Reinforcement Learning Agent that Rapidly Adapts to Multiple GridWorld Environments

(5 Marks)

Problem Statement

Design a **Meta-Reinforcement Learning (Meta-RL)** agent that can quickly adapt to different GridWorld environments. Each environment may have different obstacle positions, goal locations, reward structures, or movement conditions.

The agent should first learn **general knowledge across multiple environments** and then use a small amount of new experience to adapt rapidly to an unseen environment.

Requirements

- Create multiple GridWorld environments.
- Change obstacles and goal locations between environments.
- Define common actions such as Up, Down, Left, and Right.
- Train the agent across multiple tasks.
- Store or learn reusable knowledge from previous tasks.
- Introduce a new GridWorld environment.
- Allow the agent to adapt using a limited number of interactions.
- Compare adaptation performance with an agent trained from scratch.

Example Environments

Environment Goal Main Difference

GridWorld 1 (4,4) Fixed obstacles

GridWorld 2 (0,4) Different obstacles

GridWorld 3 (4,0) Different goal

GridWorld 4 (2,4) New obstacle layout

Expected Output

The Meta-RL agent should require **fewer episodes or steps to learn the new environment** compared with an agent that starts learning from zero.

Key Concept

Meta-RL learns how to learn, allowing an agent to transfer experience and adapt quickly to new tasks.

5. Design a Multi-Agent Reinforcement Learning (MARL) System for Cooperative Robot Path Planning

(5 Marks)

Problem Statement

Design a **Multi-Agent Reinforcement Learning (MARL)** system in which multiple autonomous robots cooperate to navigate through a shared warehouse or grid environment. Each robot must reach its assigned destination while avoiding collisions with other robots and static obstacles.

The robots should learn cooperative behavior rather than independently selecting paths that may interfere with other agents.

Requirements

- Create a grid-based warehouse environment.
- Define multiple robots with different starting and goal positions.
- Allow each robot to select actions:

- Up
 - Down
 - Left
 - Right
 - Stay
 - Define individual and shared rewards.
 - Penalize collisions between robots.
 - Penalize collisions with obstacles.
 - Reward successful arrival at the assigned destination.
 - Encourage shorter and conflict-free paths.
 - Implement communication or shared-state information when required.
- Simulate all robots simultaneously.

Example

Robot 1: Start → Storage Area

Robot 2: Start → Packing Area

Robot 3: Start → Dispatch Area

The agents must coordinate their movements when their paths intersect. **Reward Design**

Situation	Reward
Reach destination	+100
Move toward target	+5
Collision with robot	-50
Collision with obstacle	-50

Invalid movement	-10
Each unnecessary step	-1

Successful cooperative navigation +20

Expected Output

The system should demonstrate that multiple robots can learn **collision-free and efficient paths** while cooperating to complete their tasks.

Key Concept

MARL → Multiple agents → Shared environment → Cooperation/coordination → Joint decision-making → Efficient path planning

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginni ng)
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts

Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated

Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
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Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively
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